WORKED SOLUTION

STEP I 1991, Question 10

Mechanics • Equilibrium of Hanging Chains

STEP I 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.14, *Parabola, Not Catenary*: https://beyondstep.keithmcnulty.org/ch11/bridge-parabola.html

The key idea

Every joint of the chain is in equilibrium under three forces: the tensions in the two neighbouring links and the weight w hung from it. Resolving horizontally shows that every link has the *same* horizontal component of tension. Resolving vertically then says that the vertical component drops by exactly w at each joint, and since the horizontal spacing is fixed, the slopes of the links form an arithmetic progression. Slopes in arithmetic progression are the discrete fingerprint of a parabola.

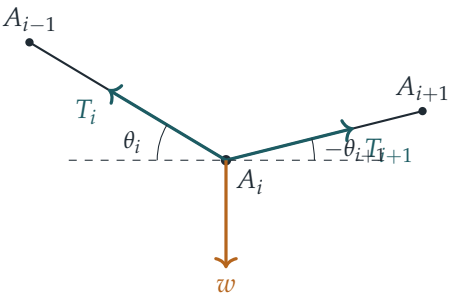
Solution

The equilibrium equations

Model the chain as light, flexible and inextensible, so the tension in each link $A_{i-1}A_i$ acts along it. Let θ_i be the angle that $A_{i-1}A_i$ makes with the horizontal, measured *downwards* as we move in the direction of increasing x, so that

$$\tan\theta_i = \frac{y_{i-1} - y_i}{h}.$$

(On the right-hand half of the bridge θ_i is negative; nothing below depends on its sign.) The joint A_i is pulled by T_i along A_iA_{i-1}, by T_{i+1} along A_iA_{i+1}, and down by the strut with force w.



Resolving horizontally and vertically at A_i, for $1 \le i \le n+1$ (indeed for every $1 \le i \le 2n+1$):

$$T_i \cos\theta_i = T_{i+1}\cos\theta_{i+1},$$
$$T_i \sin\theta_i - T_{i+1}\sin\theta_{i+1} = w.$$

These are the equations asked for.

The ratios

The first equation says that $T_i \cos\theta_i$ is the same for every link; call this common horizontal component H. It is positive, because a chain can only pull. Dividing the second equation by H and using $\tan\theta_i = (y_{i-1} - y_i)/h$,

$$\frac{H}{h}(y_{i-1} - y_i) - \frac{H}{h}(y_i - y_{i+1}) = w.$$

So if we write $V_i = H(y_{i-1} - y_i)/h$ for the vertical component of tension in link i, then

$$V_i - V_{i+1} = w \quad (1 \le i \le 2n+1), \qquad \text{so} \qquad V_i = V_1 - (i-1)w \quad (1 \le i \le 2n+2).$$

Ask yourself We have one unknown V_1 left. Which fact about the bridge fixes it?

The supports are at the same height. Adding up the drops along the whole chain,

$$0 = y_0 - y_{2n+2} = \sum_{i=1}^{2n+2}(y_{i-1} - y_i) = \frac{h}{H}\sum_{i=1}^{2n+2} V_i$$
$$= \frac{h}{H}\Big[(2n+2)V_1 - w\big(0 + 1 + \cdots + (2n+1)\big)\Big].$$

The bracket is $(2n+2)V_1 - w(2n+1)(n+1)$, so $V_1 = \frac{1}{2}(2n+1)w$. (This is also what overall vertical equilibrium suggests: the two supports share the total load $(2n+1)w$ equally.) Therefore

$$V_i = \tfrac{1}{2}(2n+3-2i)\,w, \qquad \text{that is,} \qquad y_{i-1} - y_i = \frac{(2n+3-2i)\,wh}{2H}.$$

Taking $i = n+1, n, \ldots, 1$ gives drops of $1, 3, \ldots, 2n+1$ times $wh/(2H)$:

$$\frac{h}{y_n - y_{n+1}} = \frac{3h}{y_{n-1} - y_n} = \cdots = \frac{(2n+1)h}{y_0 - y_1} = \frac{2H}{w},$$

which is the required chain of equalities. (The last numerator is printed as $(2n+1)y$ on the paper; it must be $(2n+1)h$, both from the pattern and for the ratio to be dimensionless like the others.)

The parabola

Let $k = wh/(2H)$. Summing the drops from A_0,

$$y_i = y_0 - k\sum_{j=1}^{i}(2n+3-2j) = y_0 - k\big[(2n+3)i - i(i+1)\big] = y_0 - k\,i\,(2n+2-i).$$

Writing $m = n+1$, we have $i(2m-i) = m^2 - (i-m)^2$, so

$$y_i = (y_0 - km^2) + k(i-m)^2 = y_{n+1} + k(i-n-1)^2,$$

since putting $i = m$ shows that $y_0 - km^2 = y_{n+1}$. Now $x_i = x_0 + ih$, so $(i-n-1)h = x_i - x_{n+1}$, and every point A_i, $0 \le i \le 2n+2$, satisfies

$$y = y_{n+1} + \frac{w}{2Hh}\,(x - x_{n+1})^2.$$

This is a single parabola with a vertical axis through the lowest point A_{n+1}, so the points $A_0, A_1, \ldots, A_{2n+2}$ all lie on it. The end points are included: $i = 0$ and $i = 2n+2$ both give y_0, as they should.

Checks

- With $n = 0$ there is one strut in the middle: the formula gives $V_1 = \frac{1}{2}w$, so each half of the chain carries half the load, as it obviously should.
- The result agrees with the continuous picture: a uniform load per unit horizontal length w/h hung from a cable with horizontal tension H gives $y'' = (w/h)/H$, and our parabola has $y'' = w/(Hh)$.
- In a numerical check we took 200 random bridges ($1 \le n \le 8$, random H, w, h), solved the joint equations directly as a linear system in exact fractions, and confirmed the ratios, the formula $y_i = y_0 - k\,i(2n+2-i)$, the parabola, and full force balance at every joint. For instance, with $n = 2$, $w = 2$, $H = 5$, $h = 1$ the drops are 1.0, 0.6, 0.2 on the way down, and each ratio equals $2H/w = 5$.

Common slips

- Using symmetry to say $y_n = y_{n+2}$ without comment. It is true, but it follows from the supports being level; the summing argument above proves it rather than assuming it.

- Forgetting that the horizontal component of tension is constant, and trying to find each T_i separately. The question never needs the T_i themselves (they are $\sqrt{H^2 + V_i^2}$).
- Mixing up the sign of θ_i on the rising half of the chain. Working with $\tan\theta_i = (y_{i-1} - y_i)/h$ throughout avoids this.

Beyond STEP

A chain hanging under its own weight forms a catenary, $y = c\cosh(x/c)$, but a chain carrying a load spread evenly along the *horizontal*, like a road deck, forms a parabola; this question is the discrete version of that fact. The second-difference equation $y_{i-1} - 2y_i + y_{i+1} = \text{const}$ is the finite-difference form of $y'' = \text{const}$, and the same idea underlies numerical methods for differential equations. Calculus of variations treats the catenary and its relatives. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see statics of strings and cables done carefully.

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